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How the 2026 World Cup Prediction Model Works

A complete, mathematically rigorous guide to the joint score probability engine powering every prediction on this site — from the composite team ratings through the Bivariate Poisson, market reconciliation, calibration, and edge screening.

Introduction

Predicting football is an exercise in structured humility. A single match produces roughly two or three goals from a cascade of decisions, deflections, and individual moments that no model can fully capture. Even the best football models in the world — the ones running inside sharp sportsbooks and professional syndicates — are wrong more often than they are right about any individual game.

What a good model *can* do is get the probabilities right over a large number of matches. Assign 30% to outcomes that happen 30% of the time. Be faster than the public at incorporating new information. Identify specific markets where the bookmaker's price doesn't reflect the available evidence. That is a narrower ambition than predicting winners — but it is the one that leads to long-run results.

This model produces a **full joint probability mass function (PMF) over every possible regulation-time final score** for every 2026 FIFA World Cup match. Every market you see on the site — Over/Under, Both Teams to Score (BTTS), 1X2 result, correct score — is derived mechanically from this single two-dimensional grid of probabilities. The architecture guarantees internal consistency: the numbers cannot contradict each other because they all flow from one source of truth. The model cannot simultaneously output 'Home Win = 45%' and a set of correct-score probabilities that sum to 52% across all home-win scorelines — an inconsistency that appears on sites blending data from separate models.

The pipeline runs without human intervention 24 hours a day. Six competing parametric score models are trained daily on the combined historical and 2026 tournament data. The best-performing model feeds a composite rating system built from six independent data sources. The result is reconciled against live bookmaker prices, run through an automated calibration framework, and screened for edge opportunities. Every stage is described below.

What a Joint Score PMF Actually Is

Before describing the engine, it helps to understand precisely what it produces. A PMF is a Probability Mass Function — a complete assignment of probability to every possible discrete outcome. For a football match, the **joint score PMF** is a two-dimensional grid. Each cell (h, a) holds P(Home = h, Away = a): the probability that the home team scores exactly h goals and the away team scores exactly a goals in regulation time.

The grid extends from (0, 0) through whatever maximum makes sense — typically 8 or 9 goals per team — plus an explicit tail-mass bucket for extreme scores beyond the grid boundary. Every cell, including the tail, must sum to **exactly 1.0**. This is not approximate. It is a hard constraint enforced at every stage of the pipeline. Any rounding is immediately corrected by normalizing the grid.

Here is why the grid architecture matters. Every betting market is simply a different question about this same distribution — a different way of summing certain cells. The PMF contains the answer to every possible question about regulation-time outcomes:

Market	PMF Calculation	Example (2-team match)
Over 2.5 goals	Sum all cells where h + a >= 3	Cells (0,3),(1,2),(2,1),(3,0),(2,2),(3,1),...
Under 2.5 goals	Sum all cells where h + a <= 2	Cells (0,0),(0,1),(0,2),(1,0),(1,1),(2,0)
Both Teams to Score	All cells: h >= 1 AND a >= 1	Everything except row h=0 and column a=0
Home Win	All cells where h > a	Cells (1,0),(2,0),(2,1),(3,0),(3,1),(3,2),...
Draw	All cells where h = a (main diagonal)	Cells (0,0),(1,1),(2,2),(3,3),...
Away Win	All cells where a > h	Cells (0,1),(0,2),(1,2),(0,3),(1,3),(2,3),...
Correct score 2-1	Single cell P(h=2, a=1)	One number read directly from grid
Home clean sheet	Sum entire column a=0	P(0,0)+P(1,0)+P(2,0)+P(3,0)+...

The Internal Consistency Guarantee

There is no separate model for each market type. The PMF is the entire model, and all markets are different questions about the same distribution. This guarantees internal consistency — a property that does not hold on sites that blend data from different sources or use different models for different markets.

Step 1: Rating Every Team — The Composite Prior

The first task is assigning each of the 48 teams an **attack lambda** (lambda_att) and a **defense lambda** (lambda_def). These represent expected goals scored and conceded against an average opponent at a neutral venue. The 2026 global World Cup baseline is **1.30 goals per team per match**, calibrated from the 128 matches played across the 2018 and 2022 World Cups.

No single rating system is reliable enough to trust alone. Club data does not transfer cleanly to international football. FIFA rankings can lag months behind actual form. Bookmaker odds contain genuine signal but can be distorted by public betting flow on marquee matches. The solution is a **composite prior** blended from six independent data sources, each contributing information that no other source has access to.

The Six Rating Sources

Weights shown are for matches with market odds available

Source	Primary Data	Approx. Weight	Key Strength	Primary Weakness
1. Market-Implied	Bookmaker 1X2 odds (up to 6)	30%	Late team news, sharp money	Requires available odds
2. FIFA Ranking	March 2026 points snapshot	~12%	Long-run int'l performance	Infrequent updates
3. Qualifying Record	Attack/defense efficiency	~10%	Campaign-specific form	Small sample sizes
4. Pi Rating	Goal-margin Elo (penaltyblog)	~18%	Fast form updates	Noisy on flukes
5. Elo Rating	Win/loss/draw (penaltyblog)	~35%	Stability, long-run signal	Ignores goal margins
6. Confederation Baseline	Historical WC averages by region	~5%	Prevents extreme estimates	No team-level info

Source 1: Market-Implied Strength (30% weight)

When bookmaker odds are available for a match, they encode information that no rating system can access: late team news, undisclosed injuries, sharp money from professional bettors, and the aggregate view of every serious analyst who has looked at the match. Sharp books are very good at this. Ignoring available market information is leaving real signal on the table.

The model extracts market-implied attack and defense lambdas by reverse-engineering what team strengths would produce the bookmaker's observed 1X2 probabilities. Before doing this, the bookmaker margin must be removed. Raw quoted odds always include a margin of roughly 5–8% that inflates the implied probabilities. **SHIN normalization** is applied to all odds from all bookmakers to convert them to fair (no-margin) probabilities. Using raw odds as fair probabilities would consistently bias every market comparison — this step is not optional.

Up to 6 sportsbooks contribute to each match's market picture: FanDuel, DraftKings, BetMGM, BetRivers, Caesars, and Fanatics. Their no-vig probabilities are averaged to produce a consensus view that smooths out any single book's idiosyncratic positioning.

The 30% weight is empirically determined by minimizing log-loss on completed WC2026 matches. This is the single largest weight in the composite, reflecting the genuine information content of well-supplied sharp markets.

Source 2: FIFA Ranking — March 2026 (~12% weight)

FIFA's official ranking points system, converted to an attack lambda via a calibrated sigmoid mapping. FIFA rankings capture long-run international performance across all competitions — friendlies, qualifiers, continental championships — and provide a useful signal particularly for teams with limited recent competitive data.

Primary weakness: FIFA rankings update infrequently and can reflect results from years prior. A team that had a dismal October 2025 qualifying window would see that reflected in an immediate Elo update but potentially not in the FIFA ranking until the next official release. The model uses the March 2026 snapshot — the last official pre-tournament update.

Source 3: Qualifying Record (~10% weight)

Each team's attack and defense efficiency during their qualifying campaign, Bayesian-shrunk toward their confederation's average. The shrinkage scales with the number of competitive matches played:

```
Shrinkage weight = n / (n + 3)

# where n = number of qualifying matches played
# 3 = prior strength (equivalent to ~3 'ghost' matches at the confederation avg)

A team with n=18 qualifying matches gets weight 18/21 = 0.857
A team with n=6 qualifying matches gets weight 6/9 = 0.667
```

A nation that dominated its qualifying group is genuinely more dangerous than one that scraped through on goal difference, and this source captures that signal directly from the competitive record rather than relying on rating proxies.

Source 4: Pi Rating (~18% effective weight)

The Pi rating, computed using the penaltyblog library, is a goal-margin-sensitive variant of Elo. Unlike standard Elo which updates only on match result (win/draw/loss), Pi updates on the actual score. A team winning 4-0 earns a substantially larger rating boost than one winning 1-0. This makes Pi more responsive to genuine performance shifts and better suited for a model whose primary target is predicting scores rather than just results.

Source 5: Elo Rating (~35% effective weight)

Classic international Elo, also from penaltyblog, based on win/loss/draw outcomes only. Elo is more stable over long periods than Pi precisely because it does not react to score margins — a team cannot inflate its Elo by running up the score against a weaker opponent. It serves as a stabilizing anchor that prevents the prior from swinging too sharply on an unusual scoreline, and it is the dominant signal for teams whose Pi and market data are sparse or potentially noisy.

Source 6: Confederation Baseline (~5% as a soft floor)

Historical World Cup attack averages by confederation provide the final input. These figures represent what teams from each confederation have historically scored against World Cup-caliber opponents at a neutral venue:

Confederation	Historical WC Attack Lambda	Historical WC Defense Lambda	Notes
CONMEBOL	1.45	1.10	Brazil, Argentina historically strong
UEFA	1.35	1.20	Most teams, relatively balanced
CONCACAF	1.20	1.35	Mexico strongest; field varies widely
CAF	1.10	1.30	Senegal, Morocco recent strong performers
AFC	1.10	1.35	Japan, South Korea; improving rapidly
OFC	0.90	1.45	Australia departed; small confederation

This acts as a soft floor — preventing any team from receiving a lambda estimate wildly inconsistent with what teams from their region have historically produced at World Cups. For well-rated teams with extensive data, this source contributes almost nothing to the composite. For data-sparse teams, it prevents impossible extreme estimates that would otherwise distort predictions.

Host Advantage and Altitude Adjustments

USA, Canada, and Mexico each receive a co-host adjustment of **+0.10 attack lambda** and **-0.10 defense lambda**, applied after the composite is blended. This reflects the measurable advantages of familiar geography, minimal travel, and genuine home crowd support — without overstating an effect that is smaller in a multi-host tournament than in a traditional single-host World Cup.

Venue	City	Elevation	Lambda Multiplier	Effect
Estadio Azteca	Mexico City	2,230m	0.93x	~7% scoring reduction for both teams
Estadio Akron	Guadalajara	1,560m	0.97x	~3% scoring reduction for both teams
Estadio BBVA	Monterrey	530m	1.00x (none)	No meaningful altitude effect
US venues (avg)	Various	~100m	1.00x (none)	Sea level or near-sea-level
Canada venues	Various	~50m	1.00x (none)	No altitude adjustment

Both teams' lambdas are scaled equally at altitude venues — neither side has an acclimatization advantage when both are flying in from sea level.

Tournament Adjustment: Learning from 2026 Results

Once WC2026 matches complete, results feed directly back into each team's rating via Bayesian shrinkage. The mechanism ensures the model updates on genuine tournament evidence without overreacting to a single result:

```
For a team with n completed WC2026 matches:

Shrinkage weight = n / (n + 3)
Tournament attack ratio = shrink(mean(actual_goals / prior_lambda), weight)
= weight * mean_ratio + (1 - weight) * 1.0

# Capped at +/-30%: ratio is clipped to [0.70, 1.30]

xG blend (active for all 22 adjusted teams):
blended_ratio = 0.40 * actual_goals_ratio + 0.60 * xG_goals_ratio
(when BallDontLie shot data is available; reduces noise from small sample)
```

Current Status — June 2026

As of June 2026: 22 teams have active tournament adjustments from 11 completed WC2026 matches. The xG blend is active for all 22 teams. The shrinkage mechanism means a team that played 1 match gets weight 1/(1+3) = 0.25 — only 25% of the raw adjustment survives; 75% stays anchored to the prior. A team with 3 matches gets 50/50.

Dynamic WC_AVG Scaling

This World Cup is playing fast. Historical average from 2018 and 2022: **1.30 goals per team per match**. Through 11 completed WC2026 matches, the observed rate is **1.455 goals per team per match**. This difference matters: a model calibrated to 1.30 will systematically underestimate scoring in a tournament that is producing 1.455.

```
WC_AVG_scale = Observed_2026_rate / Historical_rate
= 1.455 / 1.300 = 1.119 (11.9% uplift)

After composite prior is built:
lambda_final[team] = lambda_composite[team] x 1.119

# Applied to all 52 team lambdas (48 field + 4 spare)
# Ensures global scoring expectations match the actual 2026 environment
# Recalibrated daily as more matches complete
```

This is not a fudge factor — it is a principled correction. The composite prior tells us the relative strength of each team; the WC_AVG scale tells us what absolute level of scoring this specific tournament is producing. Both pieces of information are necessary for well-calibrated predictions.

Step 2: The Bivariate Poisson — Where the Model Gets Clever

With composite team ratings in hand, the model computes the joint score distribution. This is the technically most important step — and the one where this model goes beyond the standard football forecasting approach.

The Standard Approach and Its Flaw

The textbook starting point for football forecasting is to model each team's goals as an independent Poisson random variable. The Poisson distribution has one parameter — its mean (lambda) — and gives the probability of exactly k events occurring in a fixed interval when events happen at a constant average rate. For goals in a 90-minute match, it is a reasonable approximation: goals are relatively rare and the time between goals is roughly exponentially distributed.

Given the estimated lambdas for home and away teams, the independent Poisson joint probability is:

```
P(H=h, A=a) = P(H=h) x P(A=a)

= [e^(-lambda_h) x lambda_h^h / h!]
x [e^(-lambda_a) x lambda_a^a / a!]

# This is the product of two independent Poisson PMFs.
# Simple, fast, and well-understood. And subtly wrong.
```

The flaw is the independence assumption. **Goals are not independent.** When the first goal goes in at the 25th minute, everything changes. The losing team pushes forward, leaving more space behind. The winning team may sit deeper, inviting pressure. Tactical changes, substitutions, and psychological effects all reshape the match. The result: the correlation between home and away goals in football is small but *real and positive*. Both teams tend to score more in high-intensity, open matches. The independent Poisson misses this and systematically misprices draws relative to decisive results in certain match conditions.

The Bivariate Poisson: Three Latent Processes

The Bivariate Poisson model (Karlis and Ntzoufras, 2003) handles this properly through an elegant mathematical construction. Instead of two independent processes, the model posits that the goals in a match arise from **three independent Poisson processes**:

```
Z_1 ~ Poisson(lambda_1) : home team's independent goal process
Z_2 ~ Poisson(lambda_2) : away team's independent goal process
Z_3 ~ Poisson(lambda_3) : shared latent 'match intensity' process
(goals that arise from both teams' engagement
in a high-energy, open match)

Observed goals:
H = Z_1 + Z_3 (home total = own goals + shared intensity goals)
A = Z_2 + Z_3 (away total = own goals + shared intensity goals)
```

The intuition for Z_3: some matches have an unusually open, attacking character — both teams committing forward, creating chances, and scoring. Other matches are tight, defensive, and low-scoring regardless of the individual teams' quality. The shared intensity parameter lambda_3 captures this match-level factor that simultaneously lifts both teams' scoring rates.

The Joint Probability Formula

The joint probability of observing h home goals and a away goals under the Bivariate Poisson is derived by marginalizing over the possible values of the shared component Z_3 = k:

```
P(H=h, A=a) = e^(-(lambda_1+lambda_2+lambda_3))
x SUM{k=0 to min(h,a)}
[ lambda_1^(h-k) / (h-k)! ]
x [ lambda_2^(a-k) / (a-k)! ]
x [ lambda_3^k / k! ]

# The sum runs from k=0 (no shared goals) to k=min(h,a) (maximum possible
# shared contribution without exceeding observed totals for either team).

Key property: when lambda_3 = 0
-> only the k=0 term survives
```

```
-> formula reduces exactly to independent Poisson
-> Bivariate Poisson IS independent Poisson when lambda_3 = 0
```

The covariance between home and away goals under this model is exactly lambda_3:

```
Cov(H, A) = lambda_3
Corr(H, A) = lambda_3 / sqrt((lambda_1+lambda_3) x (lambda_2+lambda_3))
# Both are positive when lambda_3 > 0.
# This is the theoretical formalization of 'open matches produce more
# goals for both teams simultaneously.'
```

Current Calibrated Value
Calibrated value from 11 completed WC2026 matches: lambda_3 = 0.170. Positive — this World Cup is showing exactly the mutual intensity correlation the model was built to capture. Cov(H, A) = 0.170. The independent Poisson assumption significantly understates this correlation.

The Six Parametric Competitors

The model does not simply run the Bivariate Poisson. It runs **six competing parametric models daily**, trained on the combined historical and 2026 data, and selects the winner by negative log-likelihood on held-out WC2026 match results. The competition ensures the model specializes to whatever structure the 2026 tournament is actually showing:

#	Model	Core Mechanism	Key Parameter	Current Status
1	Independent Poisson	Goals independent; joint prob = product	lambda_h, lambda_a	Baseline
2	Dixon-Coles	Low-score correlation correction via rho	rho (typically < 0)	Competing
3	Bivariate Poisson	Three-process shared intensity model	lambda_3 = 0.170	LOG-LOSS CHAMPION
4	Weibull Copula	Weibull marginals linked by copula	Shape/scale params	Competing
5	Negative Binomial	Overdispersion: variance > mean	r (overdispersion)	Competing
6	Zero-Inflated Poisson	Excess zero mass beyond Poisson	pi (zero inflation)	Competing

All six compete on the same held-out WC2026 data in a walk-forward backtest — predicting each match from what was known before it was played. The model with the lowest negative log-likelihood wins and feeds all upstream calculations. The winner is reselected **daily** as more matches accumulate. Currently the Bivariate Poisson is dominant.

Step 3: Market Reconciliation (SLSQP Optimization)

Even the Bivariate Poisson model with perfect team ratings cannot know about a starting goalkeeper injury announced 90 minutes before kickoff. Market reconciliation is the mechanism that incorporates this class of information — the kind of real-time intelligence that sharp bookmakers absorb instantly and that no parametric model can anticipate.

For each match, the model extracts fair market probabilities from all available bookmakers (after SHIN normalization) across every available market type: 1X2, Over/Under totals from 0.5 through 6.5 goals, Both Teams to Score, Draw No Bet, Double Chance, and correct score lines. Up to 6 bookmakers per market, averaged after SHIN normalization to remove each book's individual margin.

A constrained optimization algorithm — **SLSQP (Sequential Least Squares Programming)** — then adjusts the parametric PMF grid to satisfy these market constraints while minimizing Kullback-Leibler divergence from the original parametric distribution:

```
Optimization problem:

minimize KL(P_adjusted || P_parametric)
= SUM{h,a} P_adj[h,a] * log(P_adj[h,a] / P_param[h,a])

subject to: P_adj[h,a] >= 0 for all (h,a)
SUM{h,a} P_adj[h,a] = 1.0
SUM{cells in market c} P_adj[h,a] = market_prob(c)
for each market c in available markets
```

KL divergence measures how far the adjusted distribution has moved from the starting point — minimizing it ensures the optimizer moves only as far from the parametric prior as the market evidence strictly requires. When SLSQP converges to a better solution than a simple weighted-average blend of the two distributions, SLSQP is used; otherwise the blend is used.

Current Performance
SLSQP performance: Only 1 non-convergence observed in 63 matches (98.4% convergence rate). When SLSQP does not converge, the weighted-average blend with 30% market weight is used as fallback. SLSQP dominates for virtually all matches where market and parametric probabilities differ materially.

Step 4: Calibration (Temperature Scaling)

A model is calibrated when its stated probabilities match observed frequencies. When it says 30%, outcomes should occur 30% of the time. Calibration is *distinct* from accuracy: an overconfident model can be directionally right (higher-probability outcomes happen more often than lower-probability ones) but numerically wrong (the stated probabilities are too extreme).

The model uses **temperature scaling**: a single scalar parameter T that adjusts the entire distribution uniformly. T is fitted by minimizing exact-score log loss on out-of-sample predictions from the 2018 and 2022 World Cups — never on training data.

```
For each cell (h, a) in the PMF grid:

p_calibrated[h,a] proportional to p_raw[h,a]^(1/T)

then renormalized: p_calibrated[h,a] /= SUM{h,a} p_raw[h,a]^(1/T)

T > 1 : exponent (1/T) < 1 -> flattens distribution (corrects overconfidence)
T < 1 : exponent (1/T) > 1 -> sharpens distribution (corrects underconfidence)
T = 1 : identity transform -- no calibration change

# Current calibrated value: T = 1.089
# Interpretation: raw model is mildly overconfident.
# The most likely scores get trimmed slightly; less likely scores get a boost.
```

Five calibration metrics are evaluated on out-of-sample data only:

Metric	Formula	What It Measures
Exact-score NLL	-SUM log(p[actual score])	Primary: how surprised by actual scores
Ranked Probability Score (RPS)	SUM (CDF_pred - CDF_actual)^2	1X2 market calibration
Brier Score	Mean (p_pred - outcome)^2	Binary market accuracy (BTTS, O/U)
Expected Calibration Error (ECE)	Mean |freq - prob| per bin	Direct probability-frequency alignment
Ignorance Score	-SUM log2(p[outcome])	Information content of predictions

Current Calibration Temperature

Current T = 1.089. The correction is modest — the raw model is reasonably well-calibrated before adjustment. An honest caveat: 128 World Cup matches is a small calibration dataset. As the 2026 tournament adds completed matches, T will be re-estimated on a growing sample.

Step 5: Edge Screening and Kelly Sizing

With a calibrated PMF for each match, the model compares its probability estimates against the bookmaker's no-vig prices for every available market. An edge exists when the model assigns higher probability than the market implies:

```
Edge = (Model probability - Market implied probability)
/ Market implied probability

# An edge of 0.08 means the model estimates the outcome is 8% MORE likely
# than the market implies -- proportionally, not in absolute percentage points.

# Example: Market implies 40%; model says 43.2%
# Edge = (0.432 - 0.400) / 0.400 = 0.08 = 8%
```

Raw edge alone is not enough to flag a market as a value opportunity. The model applies **three simultaneous filters** — all three must pass:

Filter	Condition	Purpose
Edge threshold	Edge >= 4%	Minimum signal above estimation noise; markets below this likely reflect model uncertainty rath
CI lower bound	Lower bound of 90% CI > market implied	The CI is computed by perturbing lambda +/-12%. If even the pessimistic scenario (lambdas sh
Liquidity filter	Market implied > 2%	Very low-probability markets have thin liquidity and high variance on edge estimates; excluded

Kelly sizing for flagged markets:

```
Full Kelly fraction: f* = Edge / (Decimal odds - 1)
Half-Kelly: f = f* / 2 [default]
Quarter-Kelly: f = f* / 4 [conservative]
Hard cap: f = min(f, 0.05) [5% of bankroll]

# Half-Kelly is the default because +/-12% lambda uncertainty is a fixed
# prior assumption. When true uncertainty exceeds 12%, Full Kelly overbets.
# The 5% hard cap provides a backstop regardless of the computed fraction.
```

Step 6: Closing Line Value (CLV) — The Edge Litmus Test

Counting wins and losses is a poor way to evaluate a prediction model. A model that goes 60% on 1X2 bets might simply have been backing heavy favorites. A model that goes 48% might be consistently finding genuine value on underdogs and still be profitable. Wins and losses don't distinguish these cases.

CLV is the industry-standard measure of whether a model has genuine edge. The closing line is the bookmaker's final no-vig probability immediately before kickoff — the market's best collective estimate of the true probability, having absorbed every piece of publicly available information. It represents the aggregated opinion of every sharp bettor, quantitative model, and market maker who has looked at the match. It is the most informed number available.

A model that consistently predicts *higher* probability than where the market ultimately closes is providing information the market was slow to price in. That positive CLV is evidence of genuine edge — not because the model was necessarily right about the specific outcome, but because it was ahead of the market's information flow.

A model that consistently *loses* to the closing line — predicting lower probability than where the market closes — means the market was consistently smarter, and the model's apparent positive edges were illusions driven by stale or insufficient information.

> **CLV Tracking Scope**
> 15 markets tracked per match: the three 1X2 outcomes, BTTS Yes/No, Over/Under at 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, 6.5, and Under at 1.5, 2.5, and 3.5. Closing odds are captured automatically at T-3 minutes before each kickoff by a dedicated pre-match pipeline watcher that sleeps until exactly 3 minutes before the scheduled kickoff, then fetches and normalizes closing lines across all 15 markets.

Key Terms and Definitions

A complete reference for every term, symbol, and metric used throughout the model documentation and the prediction pages.

Term / Symbol	Precise Definition
lambda (general)	The rate parameter of a Poisson distribution. Equal to both the mean and variance of the distribution. In this model: t
lambda_att	A team's attack lambda — expected goals scored against an average opponent at a neutral venue.
lambda_def	A team's defense lambda — expected goals conceded against an average opponent at a neutral venue.
lambda_h	Match-specific home team expected goals, after applying the composite prior, WC_AVG scale, and opponent adjust
lambda_a	Match-specific away team expected goals (same calculation).
lambda_1, lambda_2	Bivariate Poisson components: home and away independent goal processes after separating out the shared compo
lambda_3	Bivariate Poisson shared intensity parameter. Cov(H,A) = lambda_3. Currently 0.170 for WC2026.

PMF	Probability Mass Function. The full grid of P(h,a) values for all integer score combinations. Sums to exactly 1.0.
Joint PMF	The two-dimensional PMF grid: P(H=h, A=a) for all (h,a). The core output of the model.
Marginal PMF	The one-dimensional distribution for a single team's goals, derived by summing the joint PMF across the other team
Temperature T	Calibration parameter. p_cal ∝ p_raw^(1/T), then renormalized. T=1.089 currently. T>1 flattens (corrects overconfide
No-vig probability	Bookmaker odds with the bookmaker margin removed. Computed via SHIN normalization.
SHIN normalization	A method to remove bookmaker margin from quoted odds that assumes the margin is applied multiplicatively. Stand
KL divergence	Kullback-Leibler divergence. Measures how much one probability distribution differs from another. Minimized in SLS
SLSQP	Sequential Least Squares Programming. A constrained nonlinear optimization algorithm used for market reconciliati
Edge	(Model prob - Market prob) / Market prob. Positive = model assigns higher probability than market implies.
CLV	Closing Line Value. Model probability minus closing (pre-kickoff) no-vig market probability. Positive = model was ahe
Kelly fraction f*	Edge / (Decimal odds - 1). The bet size fraction that maximizes long-run bankroll growth rate.
NLL	Negative Log-Likelihood. -SUM log(p[actual outcome]). The primary metric for selecting among the six model compe
RPS	Ranked Probability Score. SUM (CDF_predicted - CDF_actual)^2. Standard calibration metric for ordered categorica
Brier Score	Mean squared error on binary probability forecasts. Range [0,1]. Lower is better.
ECE	Expected Calibration Error. Mean |observed frequency - stated probability| across probability bins. Directly measures
WC_AVG scale	Dynamic multiplier applied to all team lambdas to match observed 2026 tournament scoring rate. Currently 1.119.
Bayesian shrinkage	Pulling an estimate toward a prior value by a factor n/(n+k). Prevents overreaction to small sample sizes.
Dixon-Coles rho	Correlation correction parameter for low-scoring outcomes (0-0, 1-0, 0-1, 1-1). Negative rho boosts these scorelines

Worked Example: How P(2-1) Gets Calculated

To make the pipeline concrete, here is a step-by-step example of how the model computes the probability of a specific final score — say, a 2-1 home win — for a hypothetical WC2026 match.

Starting inputs (after composite prior and WC_AVG scaling):

```
Home team: lambda_att = 1.52, lambda_def = 1.10
Away team: lambda_att = 1.20, lambda_def = 1.25

Match lambda parameters (after reconciliation):
lambda_h = home_att x away_def x WC_scale = 1.52 x 1.25 x 1.119 = 2.127
lambda_a = away_att x home_def x WC_scale = 1.20 x 1.10 x 1.119 = 1.477

Bivariate Poisson parameters after calibration:
lambda_1 = 1.957 (home independent component)
lambda_2 = 1.307 (away independent component)
lambda_3 = 0.170 (shared intensity component)
```

Calculating P(H=2, A=1) using the Bivariate Poisson formula:

```
P(H=2, A=1) = e^(-(lambda_1+lambda_2+lambda_3))
x SUM{k=0 to min(2,1)=1}
[lambda_1^(2-k)/(2-k)!] x [lambda_2^(1-k)/(1-k)!] x [lambda_3^k/k!]

Constant: e^(-(1.957+1.307+0.170)) = e^(-3.434) = 0.03240

k=0 term: [1.957^2/2!] x [1.307^1/1!] x [0.170^0/0!]
= [3.830/2] x [1.307] x [1]
= 1.915 x 1.307 x 1.000 = 2.502

k=1 term: [1.957^1/1!] x [1.307^0/0!] x [0.170^1/1!]
= [1.957] x [1] x [0.170]
= 1.957 x 1.000 x 0.170 = 0.333

Sum: 2.502 + 0.333 = 2.835

P(H=2, A=1) = 0.03240 x 2.835 = 0.0919 = 9.19%

# Fair American odds for correct score 2-1: +988 (= 1/0.0919 - 1, expressed as +odds)
```

After temperature calibration (T = 1.089), the cell is adjusted:

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p_calibrated proportional to 0.0919^(1/1.089) = 0.0919^0.9183 = 0.0938
(then renormalized across all cells to sum to 1.0)

# The calibrated probability for 2-1 is approximately 0.0938 = 9.38%
# Fair odds after calibration: approximately +966
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This single cell of 9.38% feeds into the Edge Report for the correct score 2-1 market. If a bookmaker is offering +1100 American odds on this scoreline, the no-vig implied probability is roughly 1/12 = 8.33%. The edge would be (9.38% - 8.33%) / 8.33% = 12.6% — a significant value signal that would easily pass all three filters and appear in the edge report.

The Automated Pipeline: A Living System

Every stage described above runs without human intervention, around the clock. The pipeline is not a tool that runs when someone presses a button — it is a continuously running system that treats each completed World Cup match as a new data point to incorporate.

Pipeline Cycle	Frequency	Scope
Full Retraining	Daily, 8:00 AM UTC	Fetch all API data. Retrain all 6 parametric models on combined historical + 2026 data. Rebuild c
Odds Refresh	Hourly	Capture latest odds movements from bookmakers. Update CLV tracking records. Re-run predicti
Live Snapshots	Every 2 min (9 AM-3 AM ET)	Fetch current score, match clock, and live xG data. Compute conditional PMF given current state
Closing Odds	T-3 min before each kickoff	Dedicated watcher checks schedule every 15 min. When match within 15 min, sleeps to T-3. Fet

Full Model Architecture: Data Flow Summary

The complete data flow, from raw inputs to the final edge report, is summarized below. Every stage is fully automated.

Stage	Input	Output	Key Method
1. Data Fetch	BallDontLie API, bookmaker feeds	Raw odds, match results, xG, shot data	API polling + schema validation
2. Composite Prior	FIFA, Elo, Pi, qualifying, bookmaker odds	lambda_att, lambda_def per team	Weighted blend of 6 sources
3. Host/Altitude	lambda_att, lambda_def + venue metadata	Adjusted lambdas	+/-0.10 host adj; 0.93x / 0.97x altitude
4. Tournament Adj	WC2026 completed match results + xG	Per-team adjustment factors	Bayesian shrinkage n/(n+3)
5. WC_AVG Scale	Observed 2026 scoring rate	All team lambdas x 1.119	Dynamic scale to match tournament pace
6. Model Competition	Team lambdas + historical data	Winner PMF (Bivariate Poisson currently)	Walk-forward NLL comparison of 6 models
7. Market Reconciliation	Parametric PMF + bookmaker odds	Market-adjusted PMF	SLSQP minimize KL divergence
8. Calibration	Market PMF + T = 1.089	Calibrated PMF (the final grid)	Temperature scaling p ∝ p^(1/T)
9. Edge Screening	Calibrated PMF + market prices	Value bet flags + Kelly stakes	3-filter screen: edge, CI, liquidity
10. CLV Tracking	Model probs + closing line at T-3	CLV records per market	15 markets tracked per match

Honest Limitations

No model of football should be presented without an honest accounting of what it cannot do. The following limitations are real and material.

Regulation time only. All probabilities represent 90 minutes plus stoppage time. Extra time and penalty shootouts are not modeled. For knockout matches that go to extra time, the regulation-time distribution is not the complete picture of match outcomes.

lambda_3 = 0.170 is an average, not a per-match value. The Bivariate Poisson substantially improves on the independence assumption, but the shared intensity parameter is fitted across all completed matches combined. A match in the 85th minute with a team pressing for an equalizer has dramatically different dynamics than a settled 2-0 game at minute 60. The live model's score-state multipliers handle this in real time, but the pre-game prior is still an approximation of the true match-level correlation.

11 completed WC2026 matches is a small sample. lambda_3, the calibration temperature T, and the WC_AVG scaling factor are all estimated from 11 data points as of June 2026. These carry meaningful statistical uncertainty. All three will stabilize as the tournament progresses to 48 and then 64 total matches.

Lambda uncertainty is a fixed prior, not a per-match estimate. The +/-12% used to compute confidence intervals is a conservative global assumption. For teams with extensive, recent competitive data — Brazil, France, Germany — 12% likely overstates the true uncertainty. For teams from data-sparse qualifying regions, 12% may understate it. A fully Bayesian treatment would require per-team posterior distributions that the current architecture does not implement.

Odds move between prediction and kickoff. Edge estimates are calculated when the pipeline runs. By the time you see them, the market may have moved. A +6% edge computed at 8:00 AM UTC may have narrowed or disappeared by kickoff at 3:00 PM. Always verify current odds at your book before acting on any edge signal from this site.

Positive expected value does not guarantee positive returns in any finite sample. Even a well-calibrated model with genuine edge will experience losing streaks. The Kelly sizing and CI filter are designed to manage variance over time, not eliminate it from any individual result. Football remains genuinely hard to predict. A good model makes you more informed, not omniscient.

All probabilities represent regulation time (90 minutes + stoppage time) only. Extra time and penalties are excluded. This article is for informational and educational purposes only. Please gamble responsibly.